

Determining the Target Point of the Mid-Air Pinch Gesture

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ABSTRACT

Pinching is a common gesture primarily used for zooming on mobile devices, and previous studies considered utilizing it as a mid-air gesture in AR/VR. As opposed to touch screens, there is no physical contact point between the display and the fingers in mid-air pinching, which means the positional relationship between the target point for zooming and the users' finger movement in mid-air pinching could be different from that of touch screens. In this study, we investigated the relationship in mid-air pinching to estimate the target point from the hand posture, and found that the point was significantly off towards the thumb and away from the index finger (approximately 7% offset). This finding contributes to a more accurate mid-air zooming.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction techniques—Gestural input; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality

1 INTRODUCTION

As augmented reality(AR) / virtual reality(VR) devices become popular, the significance of vision-based hand gesture control is increasing. Similar to AR/VR headsets, mid-air hand gestures can be useful in smart mirrors [2], which is a kind of AR applications, to avoid touching the mirror surface and making it dirty.

In AR/VR environments, zooming is useful to comfortably look at small texts, detailed content, or far objects. On virtually all multi-touch devices, the standard way to zoom in and out is the "pinch-to-zoom" gesture [1]. Meanwhile, various ways to zoom in and out are suggested on mid-air hand gesture devices [3, 4], including mid-air pinch-to-zoom [5]. Mid-air pinch is one-handed, easy to perform, and also familiar among smartphone users.

However, the pinch-to-zoom gesture in 3DUI causes a unique problem; misalignment between intention and input due to no physical constraint. When pinching on touch screens, a displayed picture is zoomed in as the touched points of the picture follow each of the pinching fingers [1], for example, a thumb and an index finger. Meanwhile, since there is no physical point of contact between the display and fingers in mid-air pinching, the problem of misalignment between intention and input owing to physical characteristics arises.

One of the studies suggesting mid-air pinch-to-zoom is Song et al. [5], and they seem to implement it in the following manner:

- The zoom starts when the thumb and index finger meet.
- The target point of zoom is determined based on the position of the hand.
- The zoom amount is determined by the distance between the thumb and the index finger.

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This method also has a benefit because it provides users tactile feedback at the starting point.

This method requires an algorithm to find the target point specifically between the thumb and the index finger, but the aforementioned study does not provide a concrete way for this.

In this study, we investigate the positional relationship between the intended target point and the fingers for a more natural and precise mid-air pinching algorithm.

2 EXPERIMENT

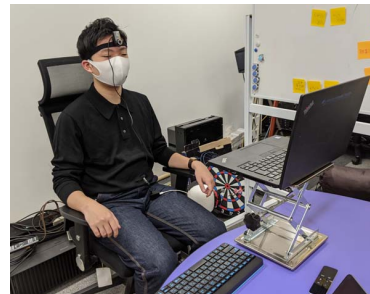


Figure 1: Experiment scene in pass-through mode. The keyboard was placed on the lap of the participant in the actual experiment.



Figure 2: The display in mirror mode (with blue point).

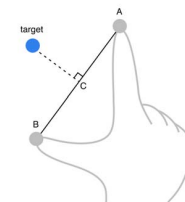


Figure 3: Definition of point A, B, and C.

We conducted an experiment in which participants performed pinch gestures in front of a camera. We developed a hand tracking application with MediaPipe (Figure 2) and performed analysis using the feature points extracted from the software. This experiment aims to find out the positional relationship between the target point and the thumb/index finger when the participant is focusing on a certain point while pinching.

A green target point was displayed on the PC. The target point turned blue when a participant made their hand pinch-shaped and their fingers were close enough to the point (Figure 2). The participant then performed a pinch-out gesture intending to zoom in on the point, and the hand posture during the gesture was recorded.

Considering applications to real-world devices, we investigated in two conditions: A) the participant sees mirrored camera vision of self like a smart mirror or AR selfie app, and B) the participant sees the vision of head-mounted camera similar to AR/VR headsets. We call these conditions mirror mode/pass-through mode for each.

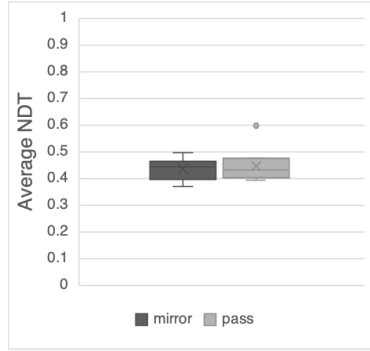


Figure 4: Average NDTs of the participant in the mirror/pass-through mode.

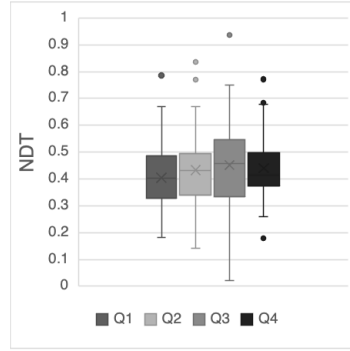


Figure 5: NDTs of the mirror mode from Quadrant 1 (Q1) to Quadrant 4 (Q4).

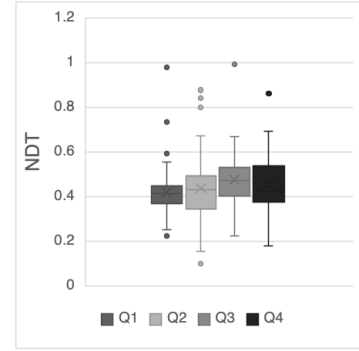


Figure 6: NDTs of the pass-through mode from Quadrant 1 (Q1) to Quadrant 4 (Q4).

We assumed that these conditions have an impact on the positional relationship based on the experience in our preliminary experiment.

We recruited 10 participants (all male in their 20s, one left-handed). First, the participants perform pinch gestures at displayed points with either the mirror/pass-through mode. For each participant, a total of 35 points are displayed, randomly selected from a 7*5 grid. When they finished their first mode, the experimenter moved the webcam and they did another mode.

For each hand posture data, we calculated the Normalized Distance from the Thumb (NDT) using the last captured frame of the pinch gesture: $NDT = BC/AB$ (see Figure 3.)

3 RESULTS

Excluding errors, we collected hand posture data of a total of 323 pinch gestures in mirror mode and 333 ones in pass-through mode.

In the mirror mode, the average NDT was 0.4368 ± 0.0394 , which means the target point was approximately 6% offset toward the thumb from the midpoint of the two fingers. T-test showed the offset is statistically significant ($p = 0.0003768, d = 2.230$). In the pass-through mode, the average was 0.4471 ± 0.0613 , which was also proven statistically significant by t-test ($p = 0.01306, d = 1.189$). Figure 4 shows the NDTs of the mirror mode and the pass-through mode. T-test indicated no significant difference between two modes ($p = 0.6511, d = 0.2057$).

We also used analysis of variance (ANOVA) to see whether the offset depends on the position of the displayed target point or not. Excluding the vertical/horizontal center, the 7*5 points were grouped in four by the position (Quadrant 1 to 4), and one-way ANOVA was conducted thereafter (Figure 5, 6). As a result, the average NDT was within the range between 0.40 and 0.48 in all groups, and ANOVA showed no significant difference between groups in both the mirror ($p = 0.2681, F = 1.322, \eta^2 = 0.0180$) and the pass-through ($p = 0.1147, F = 2.001, \eta^2 = 0.0264$) modes.

4 DISCUSSION

First, because the NDT was significantly smaller than 0.5, it was shown that when a pinch gesture is finished, the target point is off to the thumb fingertip than the index fingertip. This result is reasonable as the index finger is longer and it has a greater range of movement compared with the thumb and hence it moves more. This suggests that applications with mid-air pinch-to-zoom should zoom in on the point off to the thumb rather than the index finger to provide a more accurate zoom result as the user intended.

Second, because the ANOVA showed no significant difference between the position of the point, it was shown that we can't say the proportion of this offset depends on the relative position of the body and the hand. Therefore, in practical zooming applications, the same offset value shown above could be used in all screen region.

Third, because the t-test between the mirror and the pass-through mode showed no significant difference, it was shown that the offset does not depend on whether the mode is mirror or pass-through. This suggests that the offset is not because of the posture of the participants or how they look at their hands, but because of the nature of human hands.

In future work, we will investigate whether these findings actually contribute to the accuracy/performance and the perceived naturalness of mid-air pinching tasks. Also, more detailed observation of the gesture using 3-D hand posture scanners such as Leap Motion should be conducted.

5 CONCLUSION

To determine the position of the target point in mid-air pinching, we conducted an experiment in which participants performed pinching out from displayed points in various posture. The result showed that the target point was significantly off towards the thumb and away from the index finger (approximately 7% positional offset), irrespective of the posture of the participant or whether the person saw the screen as a mirror.

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